

Segmentation

Prof. Jungong Han (jungong.han@sheffield.ac.uk)
Department of Computer Science
The University of Sheffield

Outline

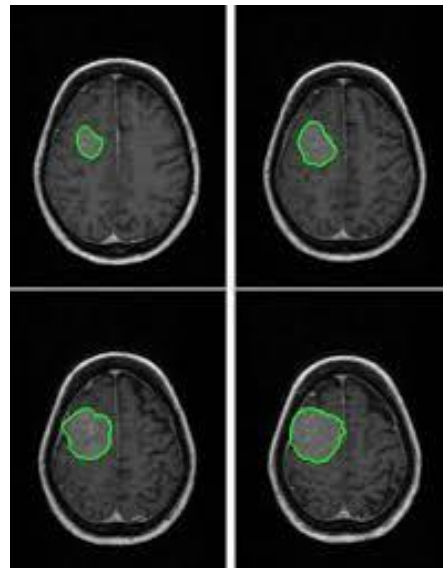
- Threshold-based Segmentation (Automatic)
- Cluster-based Segmentation
- Video Segmentation (Demos)

Segmentation

- Fundamental task in computer vision
- Finding groups of pixels that “go together” (i.e. coherent regions)
- Known statistics as cluster analysis
- Allows a simpler representation of the image (by components)
- In practice: Assign the same label to pixels that share the same characteristics

Motivation

- Scene understanding (e.g. autonomous navigation)
- Medical imaging (e.g. tumor segmentation)
- Video surveillance
- Content-based image retrieval (i.e. query by image content)
- etc...



Segmentation by Thresholding

- The most basic way of segmenting an image
- Usually, background/foreground (but not necessarily)
- Works well when the histogram distributions of each object aren't "mixed"
- Used, usually, when we can control the environment

Automatic Image Thresholding

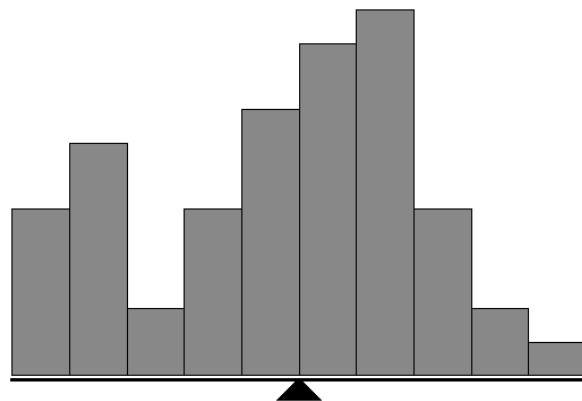
- Balanced histogram thresholding
- Otsu thresholding
- ISODATA
- Maximum entropy



Check Otsu vs. BHT: <https://youtu.be/rKWK4O4dZQ8>

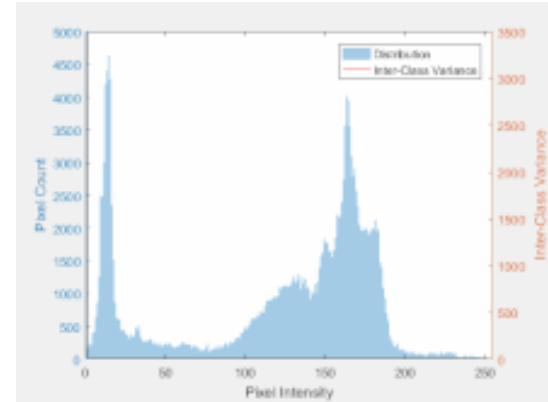
Balanced Histogram Thresholding

- Center the histogram on a weighing scale
- Repeat until no more bars in the histogram
 - Remove histogram bars from the heavier side (centering weighing scale)
 - If balance shifts, remove from the new heavier side
- Return the final center of the weighing scale



Otsu Thresholding

- For each threshold level t , calculate weighted variance of 2 classes
- Select the threshold level that minimizes the intra-class variance
- For 2 classes, it can be optimized by using the **maximum inter-class variance**



Source: Wikipedia

Otsu: Minimum of Weighted Intra-class Variance

$$w_0(t) = \sum_{i=0}^{t-1} H(i)$$

$$\mu_0(t) = \sum_{i=0}^{t-1} i \frac{H(i)}{w_0(t)}$$

$$\sigma_0^2(t) = \sum_{i=0}^{t-1} [i - \mu_0(t)]^2 \frac{H(i)}{w_0(t)}$$

$$w_1(t) = \sum_{i=t}^{\text{MAX}-1} H(i)$$

$$\mu_1(t) = \sum_{i=t}^{\text{MAX}-1} i \frac{H(i)}{w_1(t)}$$

$$\sigma_1^2(t) = \sum_{i=t}^{\text{MAX}-1} [i - \mu_1(t)]^2 \frac{H(i)}{w_1(t)}$$

$$\sigma_{\text{intra}}^2(t) = w_0(t)\sigma_0^2(t) + w_1(t)\sigma_1^2(t)$$

$$\boxed{\min_t \sigma_{\text{intra}}^2(t)}$$

Otsu: Maximum of Inter-class Variance

- From: $\sigma_{\text{intra}}^2(t) = w_0(t)\sigma_0^2(t) + w_1(t)\sigma_1^2(t)$
 - we can say that the total variance of the image is:

$$\sigma^2 = \sigma_{\text{intra}}^2(t) + w_0(t)[1 - w_0(t)][\mu_0(t) - \mu_1(t)]^2$$

- That is, the total variance is the sum of the **intra and inter-class** variances
- This allows to use the maximization of the inter-class variance:

$$\sigma_{\text{inter}}^2(t) = w_0(t)[1 - w_0(t)][\mu_0(t) - \mu_1(t)]^2$$

$$\boxed{\max_t \sigma_{\text{inter}}^2(t)}$$

BHT vs OTSU

Image



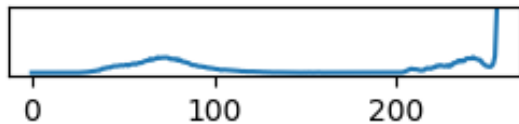
Thresholded BHT



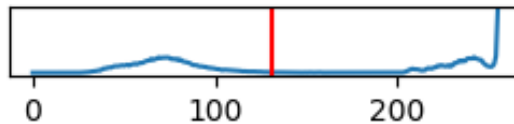
Thresholded Otsu



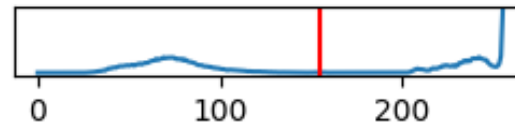
Histogram



BHT threshold: 131

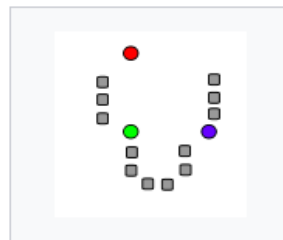


Otsu threshold: 155

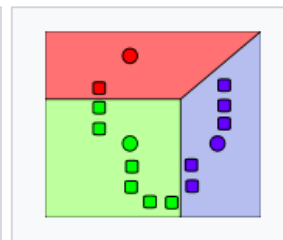


K-means Clustering: Algorithm

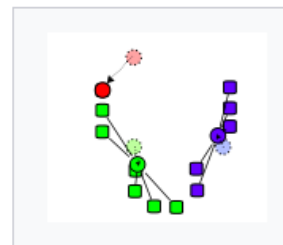
- Define the number of clusters (e.g. $k = 3$)
- Set cluster centers randomly:
- Repeat until centers do not change:
 - Assign each data point to closest center
 - Set each center to the centroid of all points it “owns”



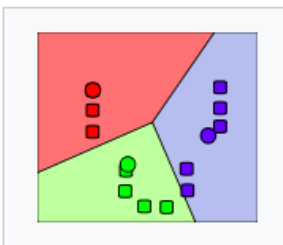
1. k initial "means" (in this case $k=3$) are randomly generated within the data domain (shown in color).



2. k clusters are created by associating every observation with the nearest mean. The partitions here represent the **Voronoi diagram** generated by the means.



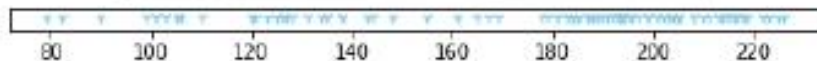
3. The **centroid** of each of the k clusters becomes the new mean.



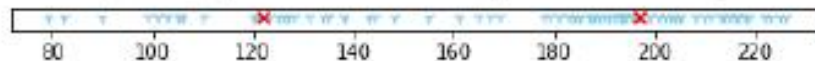
4. Steps 2 and 3 are repeated until convergence has been reached.

“Playing” with K

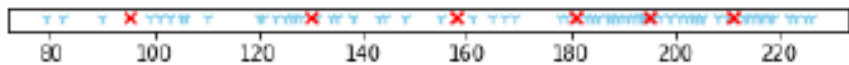
Original



$K = 2$

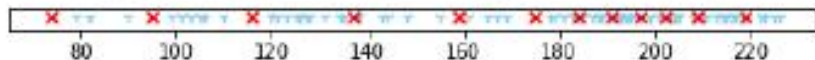


“Playing” with K

 $K = 3$  $K = 6$ 

“Playing” with K

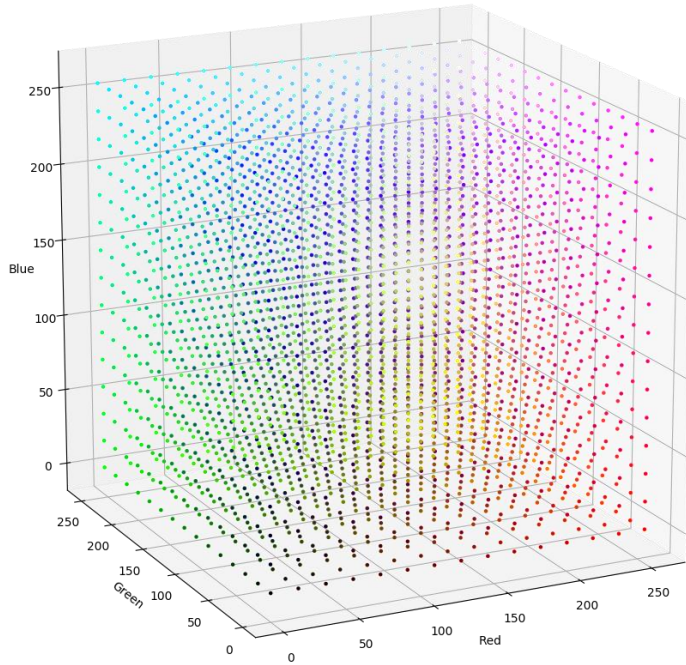
K = 12



K = 20

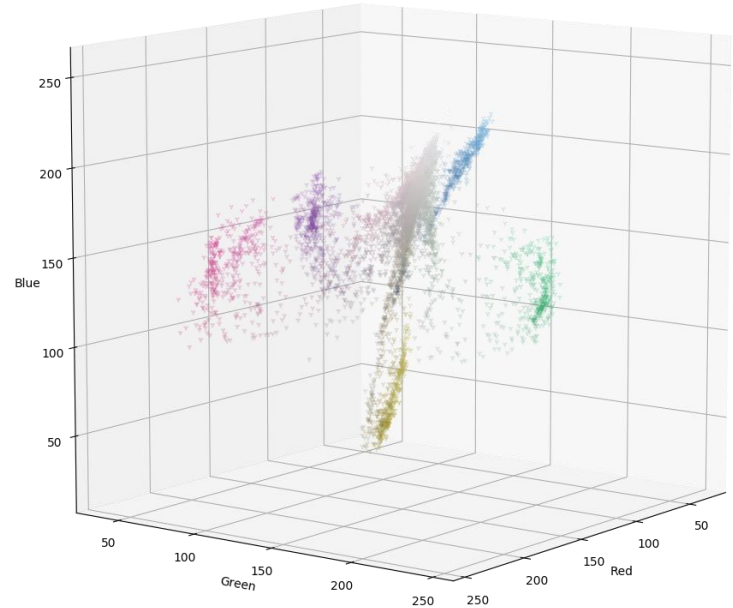


Segmenting with Color: RGB Color Space



- Instead of using intensity value only, each pixel will be represented by: (R, G, B)
- K-means algorithm can also be run in this **3D** space

Segmenting with Color: 3 features



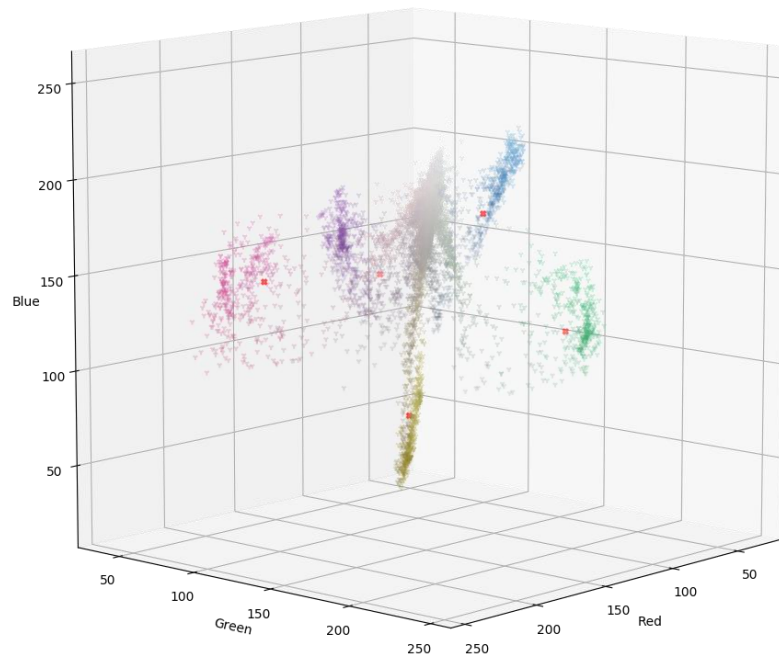
Segmenting with Color



k= 2

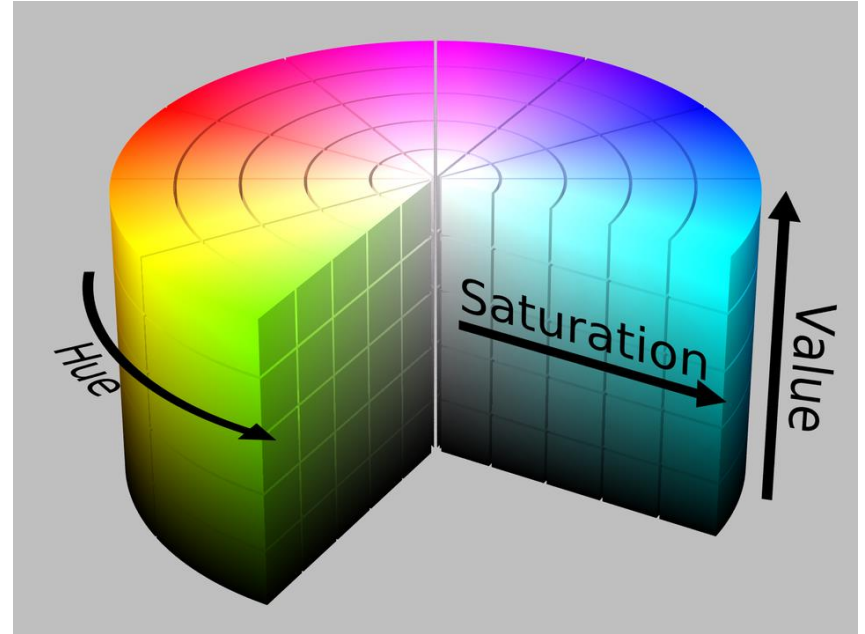


k= 7



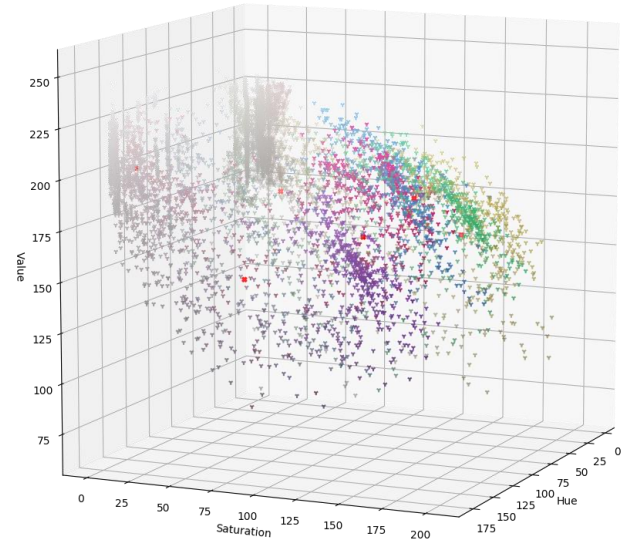
Other Color Space: HSV

- Hue; Saturation; Value
- Intuitive description of a color



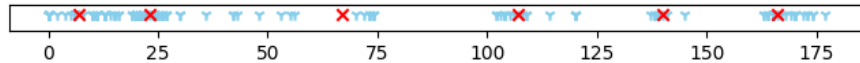
Source: Wikipedia

Segmenting with Color: HSV; k=7

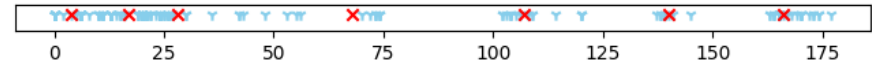


Segmenting with Color: Hue only; k=6; k=7

K = 6



K = 7



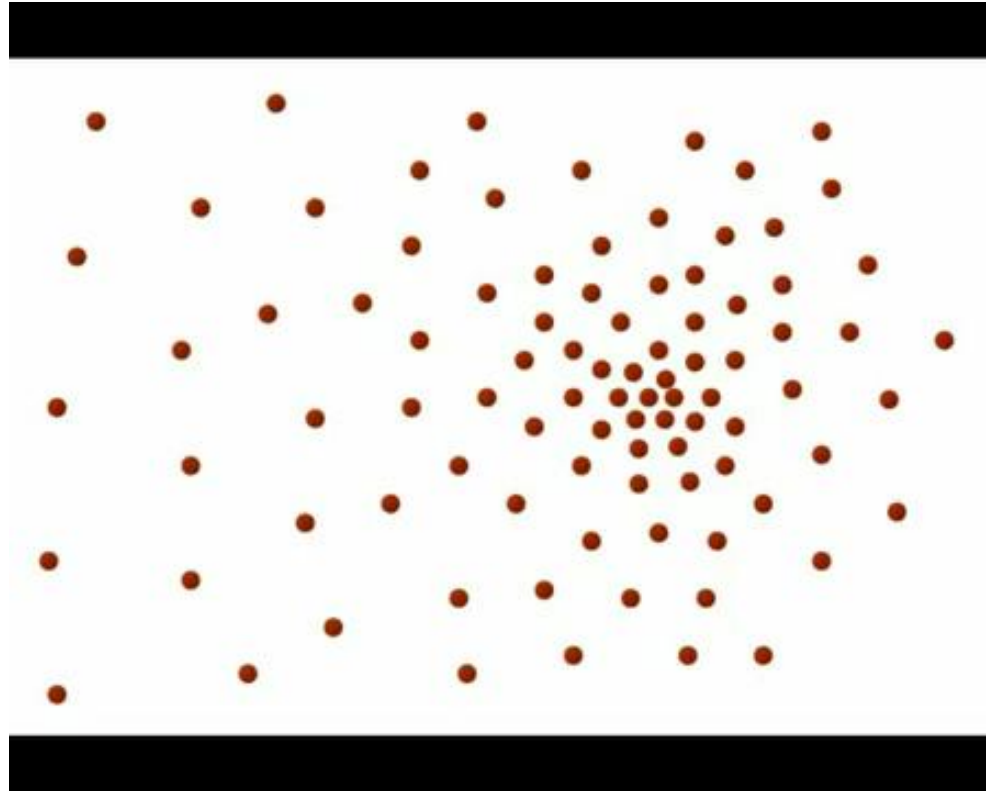
K-means: Review

- Simple to understand and implement
- Needs to know k in advance
- Computationally expensive
- Sensitive to initialization and outliers
- Expects “Gaussian/spherical-shaped” clusters

Mean Shift Segmentation

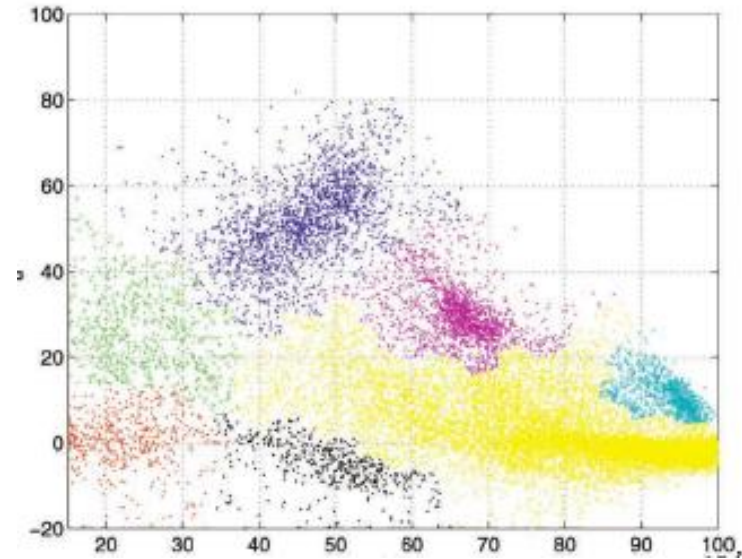
- “Searches” for modes (i.e. local maximum density) in the feature space
- Clusters are composed by all pixels “attracted” to the same mode
- Algorithm
 - Put a (Gaussian) window over each pixel
 - Move window in the direction of highest “density” until it finds the mode
 - Assign pixel to the same group of pixels whose window stops at that mode

Mean-shift: Animation (Credit: Ukainitz & Sarel)



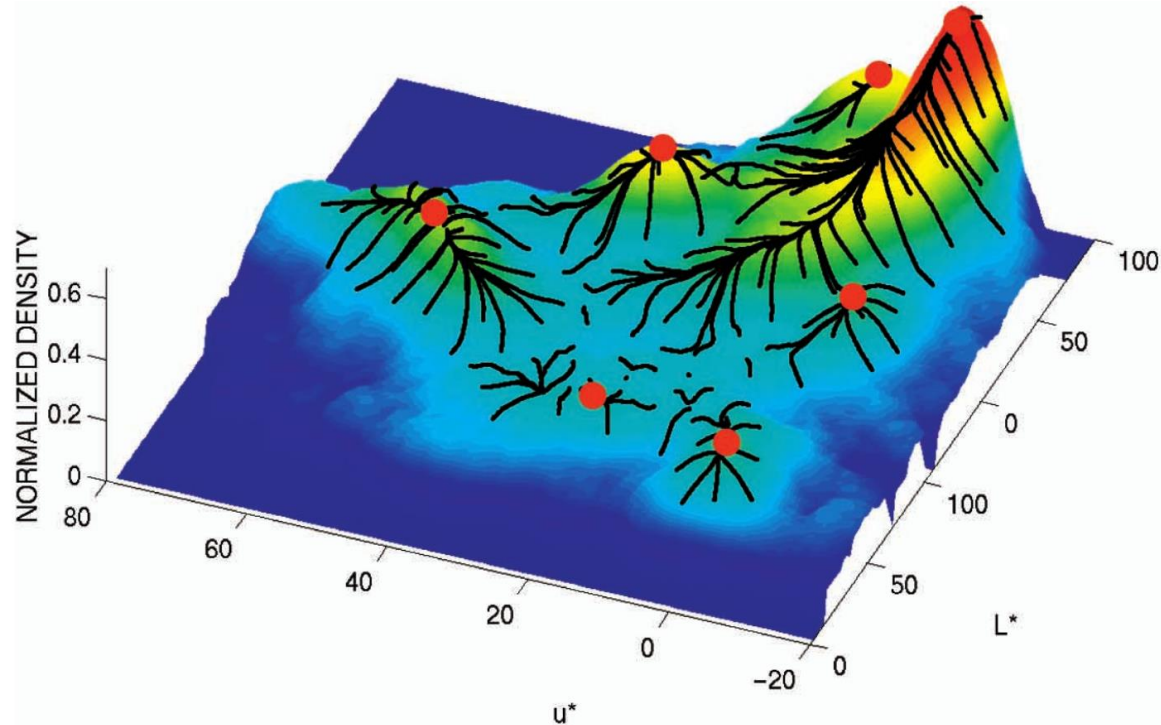
Mean Shift:

<http://dx.doi.org/10.1109/34.1000236>

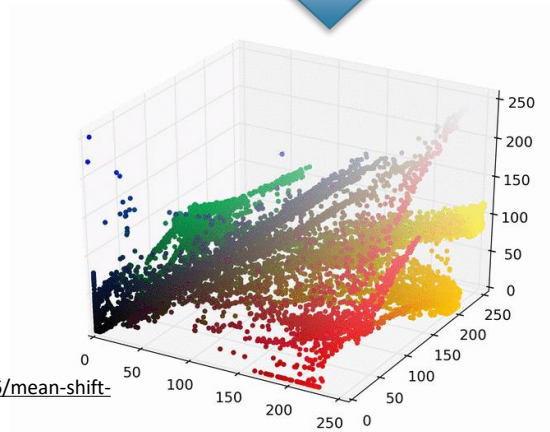
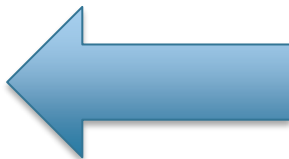
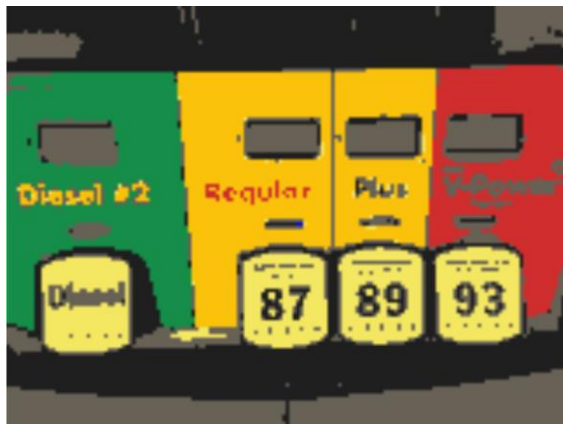
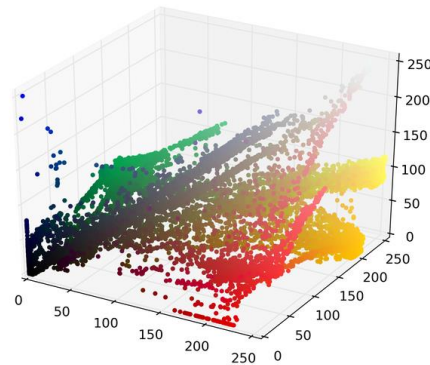


Mean Shift:

<http://dx.doi.org/10.1109/34.1000236>

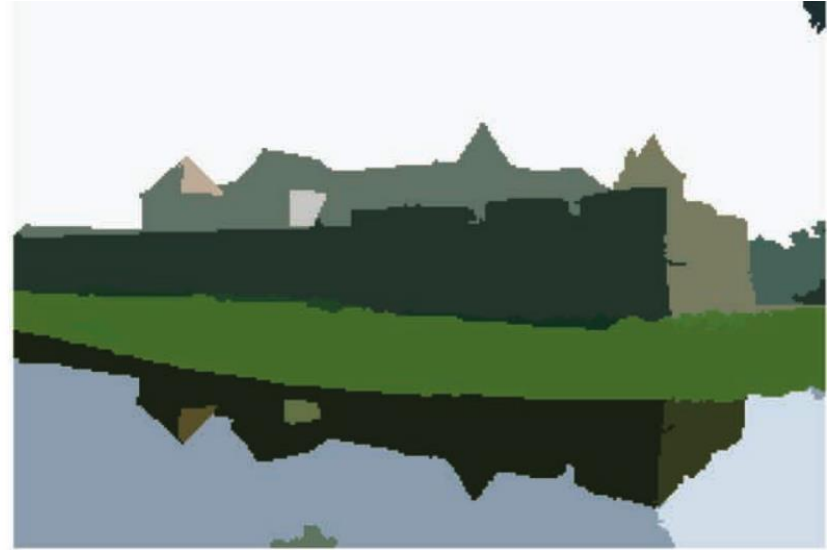


Mean Shift: Example



Source: <https://spin.atomicobject.com/2015/05/26/mean-shift-clustering/>

Mean Shift: Example 1



Mean Shift: Example 2



Mean Shift: Review

- In principle, number of clusters does not need to be known/set
- Does not assume the shape of the clusters
- 1 parameter only: bandwidth
- Setting the bandwidth is not trivial
- Computationally intensive, specially as number of features grow

Semi-supervised Video Segmentation

- Video object segmentation
 - A task of separating foreground object(s) from background on video sequences
- Semi-supervised video segmentation
 - The segmentation process is conducted under the guidance of a few given annotated frames (generally only the first frame is annotated)

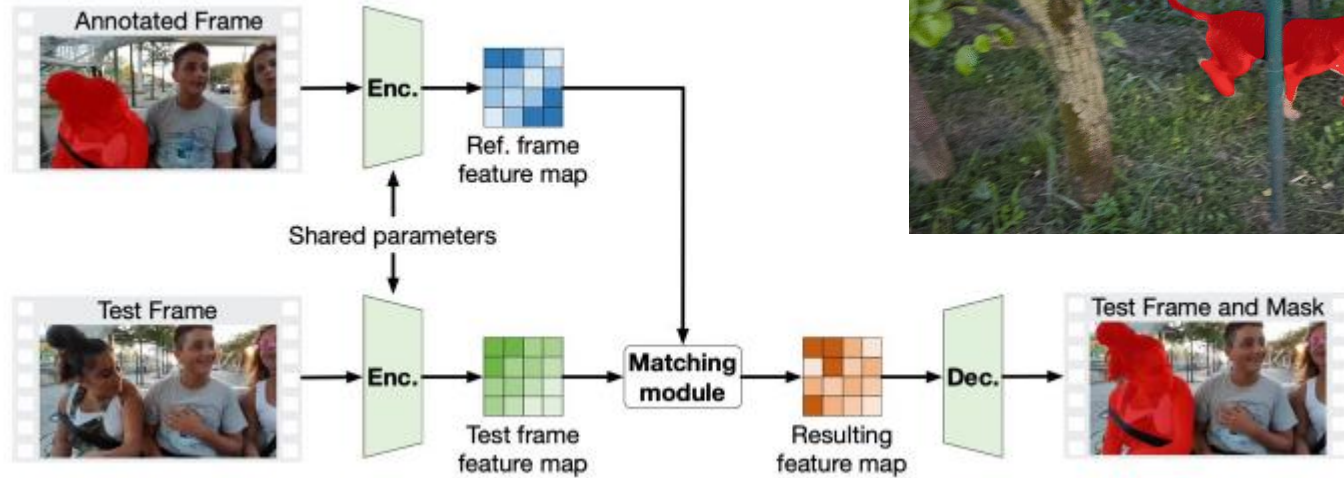


Semi-supervised Video Segmentation: Applications



video editing

Semi-supervised Video Segmentation: Method



Review

- Thresholding methods
 - Balanced Histogram
 - Otsu method
- Cluster-based Segmentation
 - K-means
 - Mean-shift
- Deep video segmentation